



MASTER IN ASTROPHYSICS
AND SPACE SCIENCE



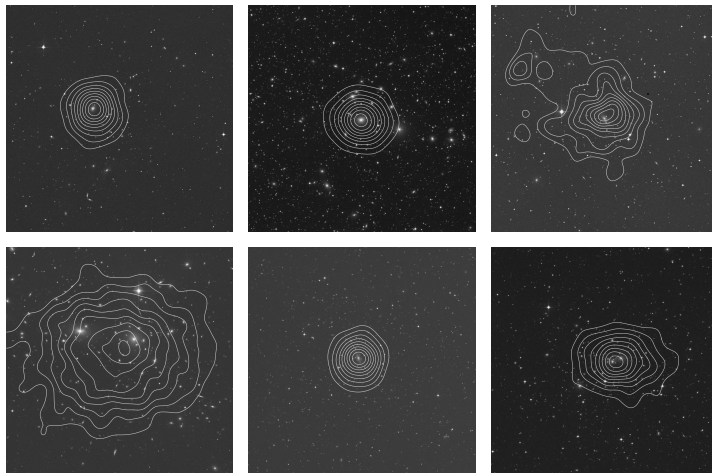
УНИВЕРЗИТЕТ У БЕОГРАДУ
UNIVERSITY OF BELGRADE

X-Ray Emitting Plasma in Galaxies and Galaxy Clusters

SPECTROSCOPY OF ASTROPHYSICAL PLASMAS

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Left to Right, Top to Bottom: Abell 85, Abell 426 (Perseus), Abell 1644, Abell 1656 (Coma), Abell 2029, and Abell 3158.

Overview

Physical Properties

- Galaxy Clusters
- Radiation Mechanisms

Radial Profiles

- Data Processing
- Theory & Results

Observations

- Detectors & Telescopes
- Spectral Characteristics
- Conclusion & Outlook

Overview

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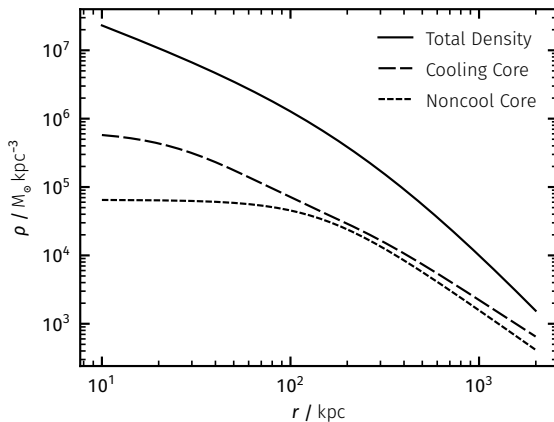
Detectors & Telescopes

Spectral Characteristics

Conclusion & Outlook

Cluster Models & Classes

- Sparse & Hot Plasma Systems
- Collisional Ionization Equilibrium (CIE)
- Purely Gravitational Baseline Assumption
- Dark Matter Halo Dominated Objects
- Selfsimilar Radial Scaling
- Deviation Defines Categories
 - Cooling Cores (CC)
 - Noncool Cores (NCC)
- Mergers Inject Energetic Shocks & Sloshing
- Central Active Galactic Nuclei (AGN) Feedback



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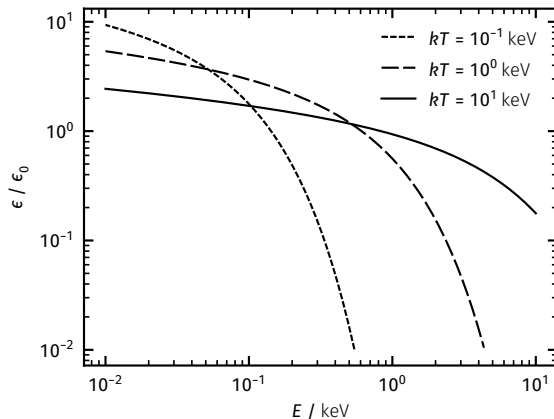
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Characteristic Emission

- Bremsstrahlung (*free-free*)
 - Broad Continuous Spectrum
- Recombination (*bound-free*)
 - Soft Continuum
 - Hard Ionization Edge
- Atomic Transitions (*bound-bound*)
 - Discrete Lines
- Sunyaev-Zeldovich Effect (*Inverse Compton*)
 - Cosmic Microwave Background Photons



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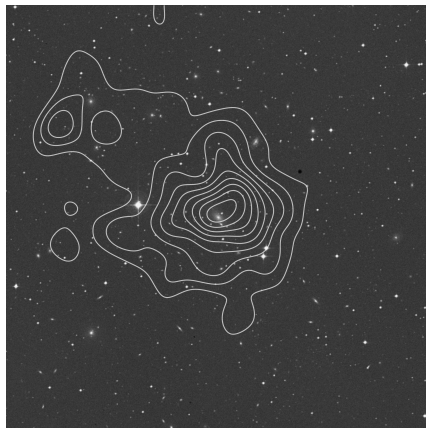
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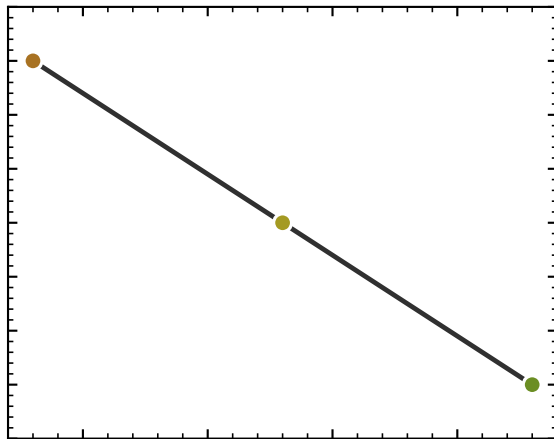
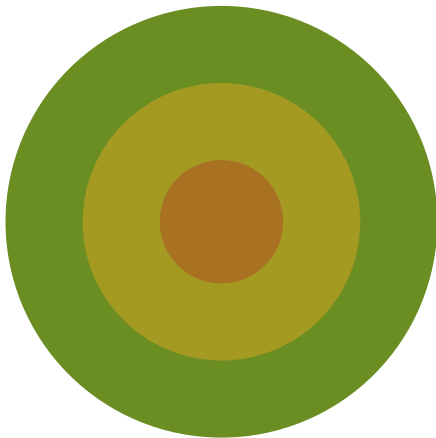
Conclusion & Outlook

Extracting Information

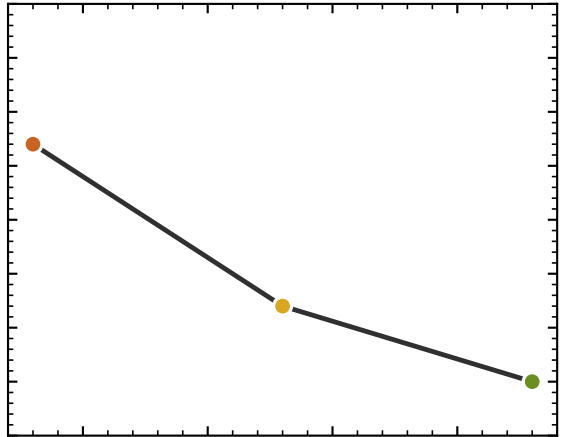
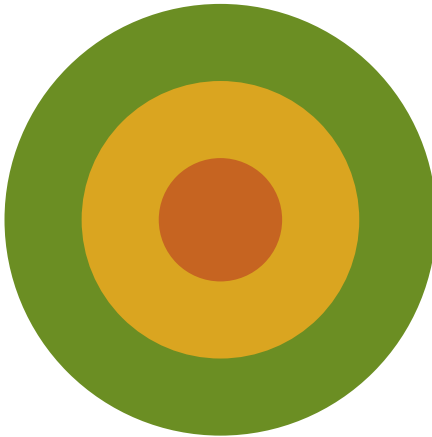
- Annulus Decomposition
- Projection Unfolding
- Spectral Fitting
 - Obtain Temperature, Density, Abundances, Redshift
 - Derive Pressure, Entropy, Mass
- Spherically Symmetric Averaging
- Disturbed Systems Breakdown Possible



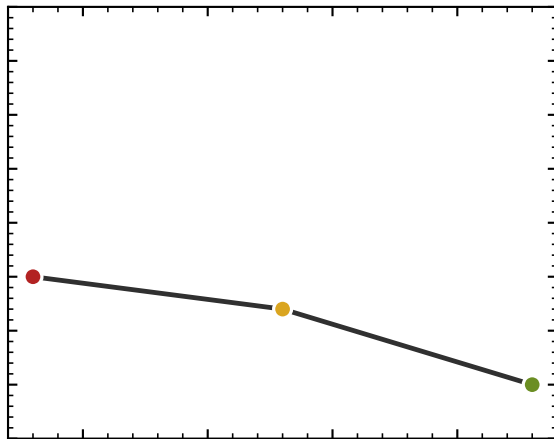
Deprojection



Deprojection



Deprojection



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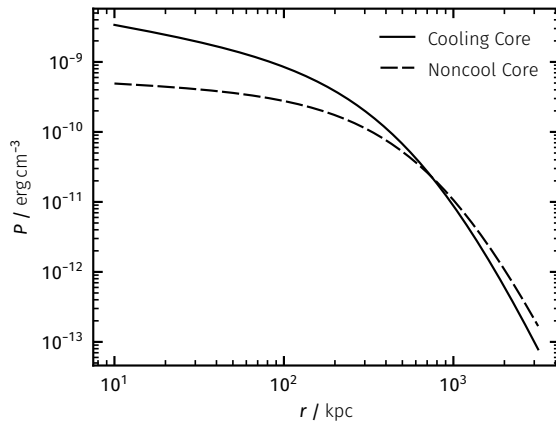
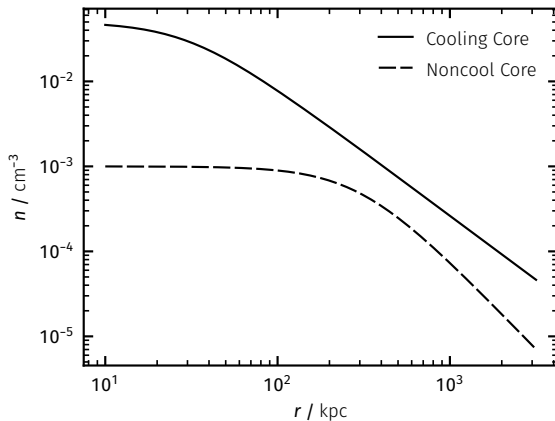
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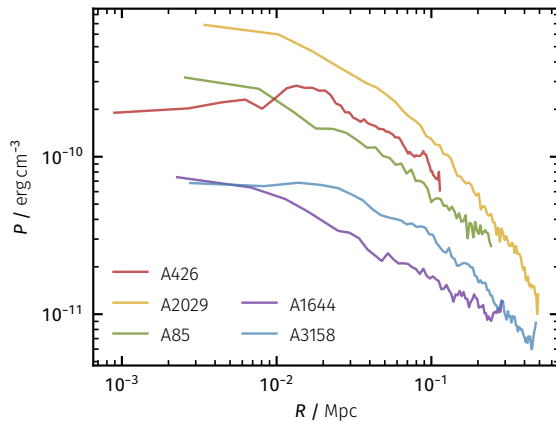
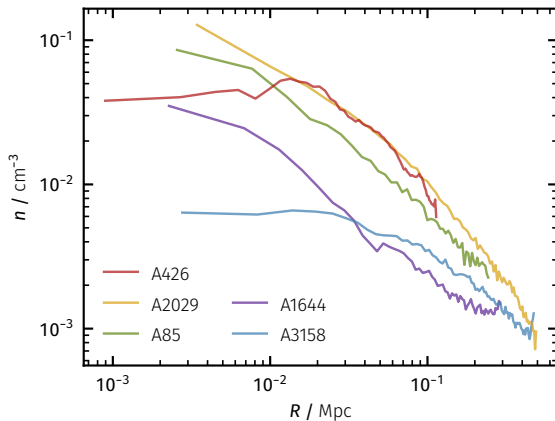
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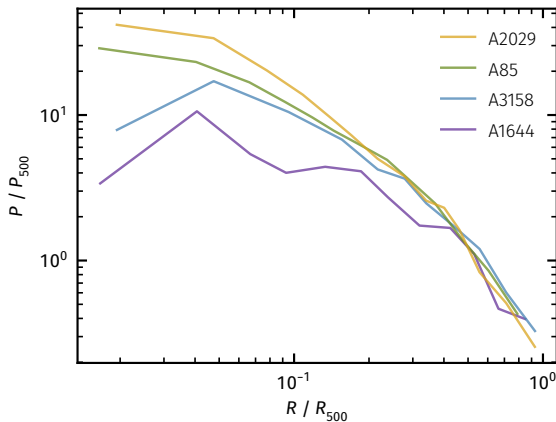
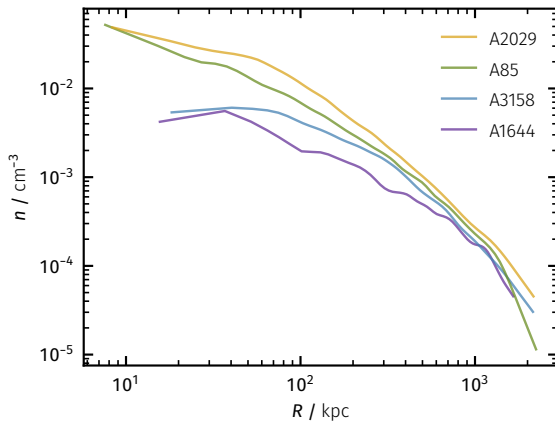
Density & Pressure



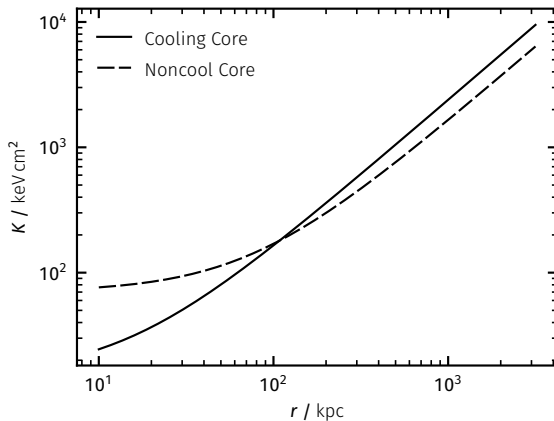
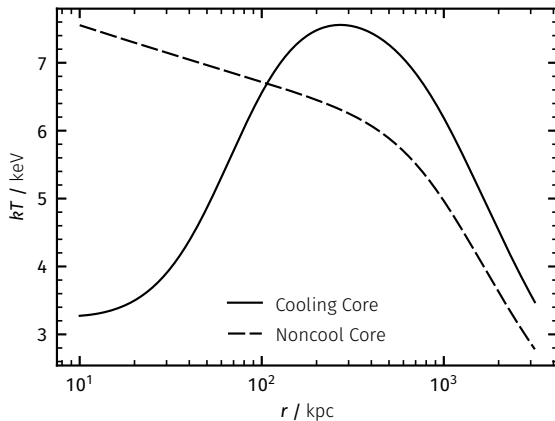
Density & Pressure



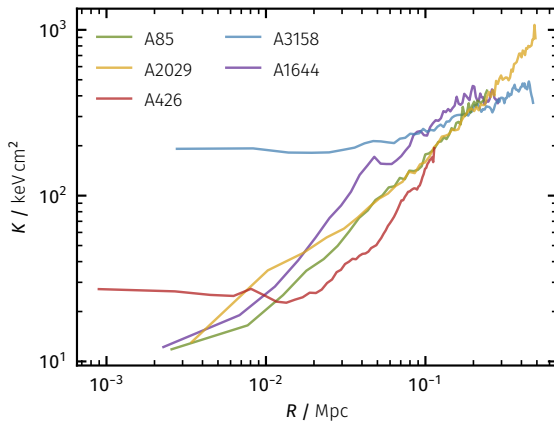
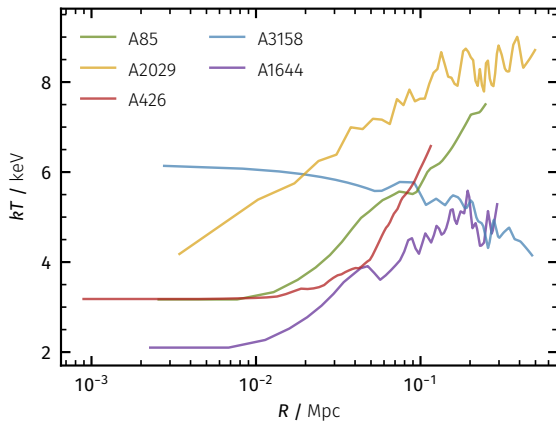
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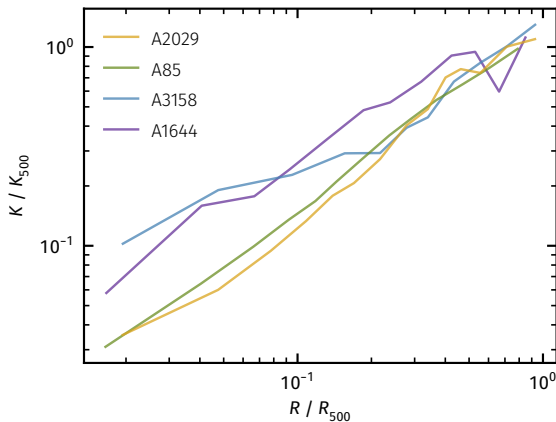
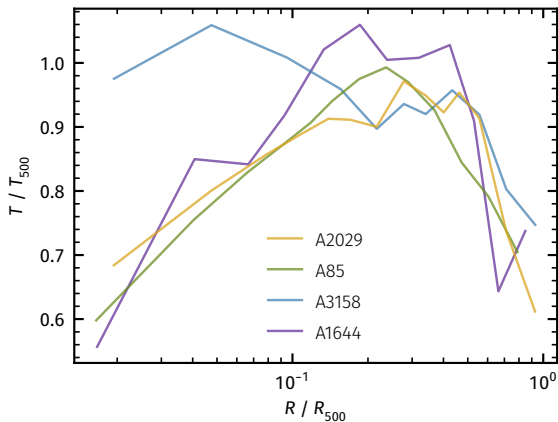
Temperature & Entropy



Temperature & Entropy



Temperature & Entropy



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Selected Missions

- Large Survey
 - ROSAT (*retired*)
 - Planck (*retired*)
 - eROSITA (*active*)
- Spatial Resolution
 - Chandra (*active*)
 - NewAthena (*planned*)
- Spectroscopic Precision
 - ASCA (*retired*)
 - XMM-Newton (*active*)
 - XRISM (*active*)



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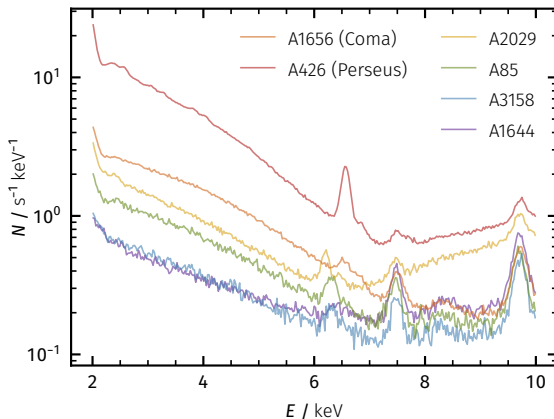
Spectrum Analysis

■ Continuum

- Bremsstrahlung Baseline
- Possible Soft & Hard Excesses
 - Multiphase Intracluster Gas
 - Warm-Hot Intergalactic Medium
 - Background Upscattering

■ Line Emission

- Powerful Flux Ratio Analyses
- Fe K & L Complexes – Thermometers
- O & Ne Lines – Supernova Tracers



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Discussion

- X-Ray Spectroscopy Enables Cosmic Structure Evolution Inference
- CC & NCC Classification Framework Traces Cluster Thermodynamical History
- Starforming Galactic Wind & AGN Feedback Stabilize Cluster Cores Against Collapse
- Highly Resolved Simulations Promise Improved Dynamics Understanding
- Future Missions Probe ICM Microproperties & WHIM Filaments
- New Simulation & Data Turbulence Comparison

Thank You!

- H. Böhringer & N. Werner, *X-Ray Spectroscopy of Galaxy Clusters: Studying Astrophysical Processes in the Largest Celestial Laboratories* (2010) *The Astronomy & Astrophysics Review*
- K. W. Cavagnolo et al., *Intracluster Medium Entropy Profiles for a Chandra Archival Sample of Galaxy Clusters* (2009) *The Astrophysical Journal Supplement*
- D. Eckert et al., *The XMM Cluster Outskirts Project* (2017) *Astronomische Nachrichten*

Resources on GitHub: [here](#)

Appendix

- Total Contained Thermal Energy

$$E_t \propto nT$$

- Radiative Cooling Rate

$$\dot{E}_c \propto n^2 \Lambda_c(T) \propto n^2 T^{-1/2}$$

- Characteristic Cooling Time

$$t_c \equiv \frac{E_t}{\dot{E}_c} \propto T^{1/2} n^{-1}$$

- Bremsstrahlung Emissivity

$$\epsilon \propto n_e n_i Z^2 T^{-1/2} g_{ff}(T, E) e^{-E/kT}$$

Appendix

■ Salpeter Initial Stellar Mass Function

$$\xi(M_*) = \frac{dN_*}{dM_*} \propto M_*^{-\alpha}$$

■ Hydrostatic Equilibrium Condition

$$\nabla P = g\rho = -\frac{GM\rho}{r^2}$$

■ Pressure Relation

$$P = nkT$$

■ Entropy Relation

$$K = \frac{kT}{n^{2/3}}$$

Appendix

■ Density Profile

$$n_{NCC} = n_0 \left(1 + \frac{r^2}{r_c^2} \right)^{-3\beta/2} \quad n_{CC} = \sqrt{n_1^2 + n_2^2}$$

■ Temperature Profile

$$T = T_0 \frac{x_{\text{cool}} + T_{\text{min}}/T_0}{x_{\text{cool}} + 1} \frac{(r/r_t)^{-a}}{(1 + (r/r_t)^b)^{c/b}} \quad x_{\text{cool}} = (r/r_{\text{cool}})^{a_{\text{cool}}}$$

■ Pressure Profile

$$P = P_0 \left((C_\ell x_\ell)^c (1 + (C_\ell x_\ell)^a) \right)^{-(b-c)/a}$$

■ Entropy Profile

$$K = K_0 + K_\ell x_\ell^\gamma \quad x_\ell = r/R_\ell$$